Coverage, agreement and provenance of candidate platinum-response labels in TCGA-OV

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Short title: Platinum-response labels in TCGA-OV

Abstract

Machine-learning studies that predict response to platinum-based chemotherapy from The Cancer Genome Atlas ovarian cancer cohort (TCGA-OV) define “response” in different, variably reported ways. In 489 platinum-treated patients we compared TCGA’s published platinum status, a platinum-free interval re-derived from the TCGA Clinical Data Resource, first-course response and, as candidate survival surrogates taken from our own earlier analyses, death before 18 or 24 months (overall-survival, OS, thresholds). Platinum status was available for 277 patients (57%). The re-derived interval agreed closely with it where both were available (κ 0.97); a patient-level audit of the 117 patients it added classed 78 as unflagged, and a provenance-filtered candidate extension that adds only those covered 355 patients (99 resistant), although it added only 2 resistant patients with RNA-seq data. At 18 and 24 months the OS thresholds flagged 21% and 39% of platinum-resistant patients; of flagged patients with a known status, 82% and 74% were resistant, against 31–32% expected without platinum information, but about half of flagged patients had no status. No threshold between 6 and 60 months agreed better than

moderately with platinum status (highest κ 0.47, at 36 months). *BRCA1/2* variants and *CCNE1* amplification pointed the same way under every label, so these markers do not resolve the comparison. Expression classifiers trained on four of the labels predicted them modestly (AUC 0.61–0.68); trained in the same 134 patients with the same folds, platinum-trained and 24-month-OS-trained models did not differ detectably in predicting platinum status (difference in AUC −0.005, 95% conditional interval −0.105 to +0.087), and refitting the whole pipeline left the comparison inconclusive. We release a versioned patient-level label table with the provenance of every re-derived call. Studies should state label definitions and class sizes, prefer platinum-specific labels for what they measure, document provenance, and treat OS thresholds as survival end-points.

Introduction

High-grade serous ovarian cancer (HGSOC) is usually diagnosed at an advanced stage and treated with cytoreductive surgery and platinum-based chemotherapy [1]. Many patients relapse with platinum-resistant disease, and no pre-treatment genomic test of platinum response is in routine use. The TCGA-OV cohort [1] has been used repeatedly to study expression-based prediction of platinum response (Table 1). A meta-analysis of machine-learning models for platinum response in ovarian cancer reported pooled concordance indices of 0.81 in training and 0.83 in validation cohorts [2], and an earlier systematic review of chemoresistance prediction found that overall survival, progression-free survival and response to chemotherapy were all used as end-points, most studies predicting survival or prognosis rather than response to treatment [3].

A classifier is trained to predict whatever its label records. Clinically, platinum response has been judged by the time since the last platinum dose, which predicts response to re-treatment [4], with a six-month cut-off conventionally defining resistance, as in TCGA's own platinum status [1]; the Gynecologic Cancer InterGroup has since replaced this dichotomy with the therapy-free interval for trial design [5], but a binary classifier still needs a dichotomous label. In TCGA-OV the candidate labels are the platinum status published with the 2011 data, a platinum-free interval re-derived from the TCGA Clinical Data Resource (TCGA-CDR) [6], the response recorded after first-line therapy and, as candidate surrogates, thresholds on overall survival (OS), the most complete outcome in the cohort. Composite groups have also been used, such as short-term survivors defined by OS of 6 months to 2 years, primary platinum resistance and progression within six months of completing platinum [7], and a review of 44 expression studies found 12 definitions of platinum resistance [8]. None of the TCGA-OV studies in our non-systematic review used an OS threshold (Table 1); we evaluate the 18- and 24-month thresholds because our earlier classifier of this cohort used them [9], not because they are established practice. Reporting guidelines require the outcome definition to be reported, and risk-of-bias tools treat it as a distinct source of bias [10,11]; in the meta-analysis above, only 4 of 39 platinum-response models were at low risk of bias in the outcome domain [2]. Each label operationalises an unobserved construct, and whether it measures that construct is separate from how well a model predicts it [12].

Earlier work re-annotated TCGA-OV treatment outcomes and related platinum sensitivity and first-line response to OS [13]. Most closely, Cocchi et al. compared platinum-free-interval categories from that re-annotation with the original TCGA annotation and reported that variability in patient selection, platinum-sensitivity definitions, data sources and analytical

pipelines limits the reproducibility of the genes reported [14]. The TCGA-CDR considers the progression-free interval generally the more informative end-point in pan-cancer analyses while recommending all four of its end-points, including OS, for TCGA-OV [6]; OS has been described as a noisy label because it includes non-cancer deaths [15]; and label choice affects drug-response prediction in cell lines [16]. In the literature we reviewed, no study had quantified agreement across platinum-based, survival-based and response-based labels in the same TCGA-OV patients, evaluated OS thresholds as tests for platinum resistance, compared what classifiers trained on each label predict, or released a patient-level label table with the provenance of each call.

We therefore asked, for training a platinum-response classifier on TCGA-OV, how the candidate labels differ (i) in how many patients they cover, which patients they call resistant and, for the re-derived interval, which records support each call; (ii) in their associations with characteristics linked to platinum response and prognosis; and (iii) in what expression classifiers trained on them predict. We do not evaluate individual studies.

Materials and methods

Data and cohort

Clinical, treatment, mutation, copy-number and expression data were obtained from cBioPortal [17,18] (PanCancer Atlas 2018 release, study ov_tcga_pan_can_atlas_2018, and the TCGA 2011 release, ov_tcga_pub) and from the TCGA-CDR [6]. Of 585 patients in the PanCancer Atlas clinical file, 496 had a treatment-timeline record, and 489 had carboplatin or cisplatin recorded as the agent; these 489 form the cohort (characteristics in S4 Table). The 89 patients without any treatment record were excluded because their treatment could not be established, and the 7

whose treatment record did not include carboplatin or cisplatin were excluded as not platinum-treated. Expression values were RNA-Seq by Expectation-Maximization (RSEM) estimates from the PanCancer Atlas release; 251 platinum-treated patients had RNA-seq data. This secondary analysis used only publicly available, de-identified data released by TCGA under its consent and governance procedures; no further ethics approval or consent was required.

Labels

Five labels were constructed for the 489 patients, each coded 1 for non-response or resistance and 0 otherwise, together with two combinations of the platinum labels (Table 2).

Platinum status. TCGA's published platinum status [1] (ov_tcga_pub, unchanged): "Platinum status was defined as resistant if the platinum free interval was less than six months and the patient had progressed or recurred" [1, Supplementary Methods S1, pp. 2–3]; sensitive tumours had an interval of six months or more, and patients without progression followed for less than six months after the last dose were excluded. Patients classed "Tooearly", with missing status or absent from the 2011 release were undetermined.

Platinum-free interval from first-line platinum. A re-derivation from different source fields: the TCGA-CDR progression-free interval time [6] minus the latest stop date of platinum regimens recorded with an adjuvant indication, both in days from diagnosis (platinum rows: treatment-timeline agent names containing "platin", which includes oxaliplatin; restricting the anchor to carboplatin and cisplatin, the cohort's agents, changed no resistant or sensitive classification among patients determinate under both rules, but left 2 patients, 1 of them among the 78 unflagged additions, undetermined). Progression less than 183 days after the end of platinum was labelled resistant and an interval of 183 days or more sensitive; patients censored before 183 days were undetermined. Patients whose recorded progression preceded the end of platinum (a

negative interval) were classed resistant (n = 8); those censored before the end of platinum were undetermined (n = 11). TCGA-CDR counts death with tumour as a progression event and dates progression by new-tumour-event records [6]. With three alternative anchors, 0 to 5 of the calls determinate under both anchors changed (κ with the primary label about 0.97), and all 78 unflagged added patients remained determinate and unchanged (S1 Text; S11 Table). A broad combined label used platinum status where determinate and this interval otherwise.

Provenance audit and combined labels. After the main results were known, we audited the records behind every re-derived call, tracing each resistant call to a recorded new tumour event (recurrence or progression) or a death counted as progression. Each patient was flagged for a redacted TCGA-CDR record, an adjuvant platinum regimen without a stop date, a later-line regimen started after the end of platinum but before the recorded event (for sensitive patients, within 183 days of the end of platinum), or an interval within 150–215 days of the end of platinum; this last flag was specified before the audit to mark calls that a dating error of about a month could reverse, and does not imply that such records are wrong. Patients with no flag were classed unflagged (clean_resistant or clean_sensitive in the label table), except that resistant calls dated only by a death were never unflagged and negative intervals formed a separate class. "Unflagged" means that no specified flag applied, not that the call was verified. The provenance-filtered combined label used platinum status where determinate and the re-derived interval only in unflagged patients (the "strict combined label" in S1 Text; column label_combined_strict). It is used in the label table and in evaluating OS thresholds against a combined reference; the other analyses use platinum status, the re-derived interval or the broad combined label. Because only the additions were filtered, it does not apply uniform quality requirements: it is a conservative extension of platinum status, and the accuracy of the added calls is not verified. Under a rule

fixed before the audit, because more than 20% of the added patients could not be classed unflagged, it became the preferred combined label and the broad label a sensitivity analysis; it is a candidate extension, not a label shown to be more accurate. The label without the 150–215-day flag (367 patients; column label_combined_strict_asrun) is a further sensitivity analysis.

First-course response. Stable or progressive disease versus complete or partial response after primary therapy, from the TCGA-CDR treatment-outcome field, then ov_tcga_pub, then the PanCancer Atlas timeline, an order fixed after the three sources were found to agree in 365 of 365 and 141 of 142 shared patients.

OS thresholds. Death before 18 or 24 months from diagnosis versus survival beyond the threshold, excluding patients censored before it. Our earlier classifier of this cohort defined platinum response as survival beyond 12, 18 or 24 months [9]; the present study supersedes that interpretation and evaluates these thresholds as candidate survival surrogates for platinum resistance, not as established practice. None appeared in the literature sample (Table 1). A 12-month threshold is reported in the Supporting information.

Coverage and agreement

Patients with and without a determinate platinum status were compared on overall survival (Kaplan–Meier medians, log-rank test), death before 18 and 24 months, the re-derived interval and first-course response (Fisher's exact test). Agreement between pairs of labels with both determinate was measured by Cohen's κ [19] with 95% intervals from 2,000 bootstrap resamples of patients, alongside the maximum κ attainable given the marginal frequencies. The 18- and 24-month thresholds were evaluated as tests for platinum resistance, with platinum status as the reference: sensitivity, specificity, positive predictive value (PPV) and negative predictive value, with Wilson 95% intervals [20]. Exploratory analyses added after internal review repeated this at

every threshold from 6 to 60 months and estimated the values expected if the OS label carried no information about platinum status (PPV equal to the prevalence of resistance; specificity equal to the proportion not flagged) and the highest sensitivity the label's prevalence allows, and bounds from assigning all OS-labelled patients without a status to resistant or to sensitive. Because patients censored before a threshold have no OS label, the measures were re-estimated with inverse-probability-of-censoring weights (IPCW; Kaplan–Meier, and a Cox model of censoring on age, stage IV and residual disease) [21] (S1 Text).

Associations with characteristics linked to platinum response and prognosis

Carriers were patients among the 272 sequenced platinum-treated patients in the 2011 release with a germline or somatic truncating or splice-site *BRCA1* or *BRCA2* variant (55 carriers, 44 with a germline variant; germline calls are from the 2011 release, because the public PanCancer Atlas mutation file is somatic-only); *CCNE1* amplification was a GISTIC call of 2 in the same release. Each label was also tested against residual disease after surgery (more than 1 cm versus less, as defined by TCGA [1], and, alternatively, any versus none), stage IV disease, Aneuploidy Score, tumour mutational burden (TMB) and age. Binary characteristics were tested by Fisher's exact test with conditional odds ratios (OR) for non-response, and continuous ones by the Mann–Whitney U test with rank-biserial correlation (r); for the five pre-specified clinical and tumour characteristics, P values were adjusted by the Benjamini–Hochberg method [22] within each label. To compare labels directly in the patients with platinum status and both OS thresholds determinate, differences in the log OR for *BRCA1/2* and residual disease and in r for aneuploidy were estimated with 95% intervals from 2,000 bootstrap resamples, adding 0.5 to every cell (Haldane correction) so that no resample was discarded. The smallest difference detectable with

80% power was derived approximately from each interval (exploratory); it describes sensitivity, not power.

Classifiers

Logistic regression (L2 penalty), random forest (300 trees) and a radial-basis-function support vector machine (SVM) were fitted with balanced class weights and fixed seeds in scikit-learn 1.9.0 [23] (analyses run in version 1.7.2 are listed in S1 Text). Expression was log2(RSEM + 1)-transformed and restricted to the 15,994 genes also measured on the external platform; the 500 genes with the highest variance and the standardisation parameters were computed on training data only.

All-patients analysis. For four training labels (rows of Fig 3A), models were trained and evaluated by five-fold stratified cross-validation repeated five times over the 251 patients with RNA-seq data, and pooled out-of-fold scores were evaluated against every label (a train-label by test-label matrix of AUCs; mean over three models and five repeats, with range). These AUCs describe which outcomes the classifiers predict, not what biological information they use; because labels differ in patients and class balance, the analysis is descriptive (its pre-specified decision rule is in S1 Text).

Controlled comparison. Added after the main results were known, this trained the same pipeline with platinum status, OS at 18 months and OS at 24 months in the 134 patients with RNA-seq data in whom all three were determinate, using the same outer folds for every label (five-fold cross-validation stratified on platinum status, repeated 20 times). The primary quantity was the difference in AUC for platinum status between platinum-trained and 24-month-OS-trained models (platinum-trained minus OS-trained), averaged over the three models, each model's difference computed from its own predicted probabilities. Two secondary differences were

computed the same way: platinum-trained minus 18-month-OS-trained models for platinum status, and 24-month-OS-trained minus platinum-trained models for OS at 24 months. Its conditional 95% interval came from 2,000 bootstrap resamples of patients, stratified by the target label, using out-of-fold scores averaged over repeats, without refitting. As a separate sensitivity analysis, the whole pipeline was refitted in 500 bootstrap samples of the 134 patients drawn within platinum-status groups, and evaluated on the patients not drawn. Because single fits on about 63% of the distinct patients estimate a different quantity from repeated cross-validation, these out-of-bag results are reported separately, with their own median and percentile range.

Exploratory analyses

Three analyses were exploratory, added after the main results were known (S1 Text). First, overall survival was divided into the platinum-free interval and survival after progression (among patients who progressed), each modelled by univariable Cox regression on age, stage, residual disease, *BRCA1/2*, *CCNE1* and Aneuploidy Score. The platinum-free interval, measured from the end of first-line platinum, included only the 408 patients at risk at that time (interval above 0 days); setting the 28 intervals of 0 days or less to 1 day was a sensitivity analysis (S10 Table). Survival after progression was overall survival minus the TCGA-CDR progression-free interval, so patients whose progression event was their death were excluded. The two models were not compared formally. A rank-based test of proportional hazards gave $P < 0.05$ in 2 of the 14 models (age for the platinum-free interval, Aneuploidy Score for survival after progression), so those two hazard ratios should be read with caution; a quadratic term detected no non-linearity for either continuous covariate ($P \geq 0.28$; S1 Text). Second, a logistic regression on age, stage IV and residual disease (> 1 cm), with five-fold cross-validation repeated five times, asked how far

each label can be predicted from clinical characteristics alone. Third, the re-derived interval was recomputed without the 38 patients whose TCGA-CDR record is marked as redacted.

Supplementary analyses

First, the TCGA-trained classifiers were transferred to an independent microarray cohort, GSE63885 [24], restricted to serous and undifferentiated tumours (80 samples). Genes were z-scored within GSE63885, so this is a batch-level transfer assessment, not a frozen pipeline applied to single new patients. Its platinum-resistance label (disease-free survival < 180 days; 34 resistant, 40 sensitive) and clinical-response label (stable or progressive disease versus complete or partial response; 10 of 74) are defined verbatim, with sample identifiers and probe mapping, in S7 Table; S1 Text reconciles these counts with the original publication and with Chen et al. Differences in transfer performance cannot be attributed to the training label alone. Second, the 10 genes of Chen et al. [25] and the 5 genes of Gao et al. [26] were used as fixed inputs to our pipeline and compared with 1,000 random gene sets of the same size drawn from the 15,484 common genes with complete data; both sets were selected in data that included TCGA-OV patients, which favours them in TCGA, and this gene-set transfer neither replicates nor tests the original models. Third, we measured the AUC inflation from selecting genes on the full dataset before splitting, a well-documented error [27–30], with OS-threshold labels and the pipeline's own simpler preprocessing (untransformed RSEM values for all 20,531 genes).

Literature review

For TCGA-OV studies of expression-based platinum-response prediction found in an initial and a targeted search [25,26,31–40], we recorded from the full text and supplements the label, its stated definition and source, the class sizes, the validation design and the reported discrimination; items not reported were recorded as "not stated", and no definition was inferred.

Neither search was systematic, and databases, dates and search terms were not recorded, so the sample is illustrative, the selection cannot be reproduced and “none” means none in this sample.

Pre-specification

Any predictions or decision rule were written into the script header before the analysis was run (exceptions are disclosed in S1 Text). The scripts for the analyses of *BRCA1/2* and *CCNE1*, of patients without a platinum status, of prognostic characteristics in the full cohort, of provenance and of the controlled classifier comparison were deposited in the public code repository before they were run (commits d5bd844, e98b3d9 and d349b2b); other headers are archived with the code but not timestamped externally. Three levels of specification apply: predictions for the initial label analyses, written before those analyses but after the author’s earlier work on the same cohort; secondary predictions, written after earlier results of this study had been seen and possibly informed by them (the five deposited analyses); and post hoc analyses with decision rules (the restricted survival decomposition and the anchor, censoring-weight, event-type and out-of-bag sensitivity analyses), alongside exploratory or descriptive analyses without predictions. No hypothesis was fixed before any inspection of TCGA-OV data, and commit timestamps establish only the order of writing and running. The descriptive full-cohort tables and the exploratory analyses were added after the main results were known. Two material deviations occurred: the provenance-audit code omitted one pre-specified flag (an interval within 150–215 days), so the reported label comes from a correction run that adds only that flag; and the out-of-bag analysis’s pre-specified rule, which made its range the main uncertainty, was not followed because it estimates a different quantity. S1 Text gives each analysis’s identifier (A1–A23), predictions and outcomes (including those that failed), software environments and development

history. Reporting follows STROBE [41], and STARD 2015 for evaluating OS thresholds as tests [42] (S2 Text).

Results

Studies label patients in different ways

Twelve studies of expression (mRNA or protein) and platinum response that used TCGA-OV, or a cohort used alongside it, applied at least five label constructions, and none used an overall-survival threshold (Table 1). Five labelled patients by progression or recurrence within six months of the end of platinum or chemotherapy, but differed in the reference point and in whether intermediate patients were excluded [25,31,33,34,36]; one compared platinum-refractory disease with no progression within two years [39]; one used progression-free survival as a surrogate [35]; and two used class counts identical to those of TCGA's published platinum status (90 resistant, 197 sensitive) without stating a definition [37,38]. Four did not fully state how their TCGA patients were labelled [26,32,37,38]. In this illustrative sample, the proportion of TCGA patients labelled resistant ranged from 19% to 44% (S1 Table).

Platinum status covers just over half of treated patients

TCGA's platinum status was determinate for 277 of 489 platinum-treated patients (57%; 87 resistant, 190 sensitive), and for 145 of the 251 with RNA-seq data (Fig 1A, Table 2). The remaining 212 were missing (87), too early to classify (57) or absent from the 2011 release (68). Among patients in the 2011 release, no overall-survival difference was detected between those without a determinate status and those with one (median 47.5 versus 47.7 months; log-rank $p = 0.37$), but the group was mixed: patients with missing status had a median survival of 39.0 months, whereas only 10 of 57 patients classed as too early had died. Within the 2011 release,

patients without a status were more often dead before 18 months (16 of 96 versus 22 of 268; OR 2.23, 95% CI 1.04–4.70) and, less clearly, before 24 months (24 of 95 versus 43 of 259; OR 1.70, 95% CI 0.92–3.09), and more often had stable or progressive disease after first-line therapy (28 of 114 versus 28 of 255; OR 2.63, 95% CI 1.41–4.91), but were less often resistant by the re-derived interval (13 of 68 versus 89 of 272; OR 0.49, 95% CI 0.23–0.96; S4 Table). The re-derived interval was determinate for 389 patients (358 without the redacted TCGA-CDR records, with almost unchanged κ with platinum status; S10 Table); coverage of the other labels is given in Table 2.

Of the 389 patients with a re-derived interval, 272 also had platinum status and 117 were added by the re-derivation (26 resistant). Of the 26 added resistant calls, 20 were dated by a recorded new tumour event and 6 by a death counted as progression (S11 Table). Of the 117 added patients, 78 were unflagged (12 resistant, 66 sensitive) and 39 were not. Counting each patient once, 1 had a negative interval, 6 had a resistant call dated only by a death counted as progression, 20 had another record problem (a redacted TCGA-CDR record, adjuvant platinum without a stop date, or a later-line regimen before the event), and 12 were flagged only for an interval within 150–215 days (Fig 1D, S11 Table); these 12 are exactly those the label without that flag adds. In all four patients in whom the two platinum labels disagreed, the re-derived interval called resistant a patient TCGA had called sensitive. One had an interval near the cut-off (154 days); the other three (18, 101 and 148 days) had a recorded recurrence and no flag, and one of these (101 days) became sensitive under two of the three alternative anchors, which themselves create other disagreements; the remaining two are not explained by the available records (S11 Table). Two of the 12 unflagged resistant additions rest on a new tumour event of unrecorded type; “unflagged” means that no specified flag applied, not that the call was verified.

Setting aside the 10 resistant calls dated only by a death (6) or by an event of unrecorded type (4, including the 2 above) changed OS-threshold sensitivity and PPV against the broad combined label by at most 0.04 and the *BRCA1/2* and *CCNE1* odds ratios by at most 0.05, and none of these comparisons changed whether its interval included 1 (S11 Table). The provenance-filtered combined label was determinate for 355 patients (99 resistant, 256 sensitive; Fig 1A); without the 150–215-day flag it would cover 367 (104 resistant). Its practical gain for expression classifiers is small: among patients with RNA-seq data it adds 22 patients to platinum status, only 2 of them resistant (145 to 167 patients; 41 to 43 resistant). S11 Table gives its OS-threshold test characteristics.

OS thresholds agree at best moderately with platinum status

Where both were available, the published and re-derived platinum labels agreed in 268 of 272 patients (κ 0.97, 95% CI 0.93–0.99; Fig 1B). Agreement of the OS thresholds with platinum status was low: κ 0.24 (0.14–0.35) at 18 months and 0.37 (0.25–0.49) at 24 months, against maximum attainable values of 0.33 and 0.59. At 18 months the OS threshold flagged 18 of 84 resistant patients (sensitivity 0.21, 0.14–0.31) and 4 of 184 sensitive patients (specificity 0.98); at 24 months it flagged 32 of 83 resistant (0.39, 0.29–0.49) and 11 of 176 sensitive patients (specificity 0.94; Fig 1C). These values are largely set by how few patients die early: with no information about platinum status, the same labels would have specificities of 0.92 and 0.83, and their prevalence allows sensitivities of at most 0.26 and 0.52. Of the flagged patients with a determinate status, 18 of 22 (PPV 0.82, 0.61–0.93) and 32 of 43 (0.74, 0.60–0.85) were resistant, against 0.31 and 0.32 expected without platinum information. However, 25 of the 47 patients flagged at 18 months and 35 of the 78 flagged at 24 months had no platinum status; if all of them were sensitive or all resistant, the PPV would lie between 0.38 and 0.91 at 18 months and

between 0.41 and 0.86 at 24 months. These values also exclude patients censored before each threshold (8 and 17 of those with a platinum status) and 1 platinum-sensitive patient who died without a recorded survival time (S2 Table). With inverse-probability-of-censoring weights, sensitivity, specificity and PPV changed by at most 0.04 (S1 Text). The Kaplan–Meier weights assume censoring unrelated to outcome, and the Cox weights unrelated given age, stage and residual disease; estimating the censoring distribution only in patients with a status, or within each status group, changed the estimates by at most 0.01. Weighting does not address missing platinum status. Across thresholds from 6 to 60 months (exploratory), sensitivity rose and specificity and PPV fell, and agreement was at best moderate: κ was highest at 36 months (0.47; sensitivity 0.64, specificity 0.83, PPV 0.65; S2 Table). The two OS thresholds agreed with each other (κ 0.71), and first-course response agreed weakly with every other label (κ 0.21–0.26).

Characteristics linked to platinum response point the same way under every label

BRCA1/2 carriers were less often labelled non-responders under every label (Fig 2A, Table 3): 7 of 37 carriers versus 54 of 149 non-carriers by platinum status (OR 0.41, 95% CI 0.14–1.04), 8 of 47 versus 64 of 179 by the re-derived interval (0.37, 0.14–0.87), 1 of 51 versus 23 of 195 by OS at 18 months (0.15, 0.004–0.97) and 3 of 50 versus 43 of 192 by OS at 24 months (0.22, 0.04–0.75); the evidence was imprecise for first-course response (0.64, 0.18–1.80). In the 180 patients with platinum status and both OS thresholds determinate, the difference in log OR between platinum status and the OS thresholds was +1.12 (95% CI −0.16 to +2.00) at 18 months and +0.95 (−0.52 to +2.71) at 24 months; the approximate minimum detectable ratios of odds ratios were about 4.7 and 10, so these comparisons do not show that the labels behave alike (S3 Table). *CCNE1* amplification was associated with non-response in the same direction under every label (OR 1.11–2.01; Fig 2B).

Residual disease and aneuploidy behaved similarly (Fig 2C and 2D, Table 3). Residual disease > 1 cm was associated with non-response under both OS thresholds (OR 2.86, 1.30–6.29, and 2.41, 1.28–4.50) and first-course response (2.55, 1.31–4.91), but not under platinum status (1.12, 0.59–2.09); any residual disease versus none was associated with platinum status (3.22, 1.40–8.35), the re-derived interval (3.04, 1.44–7.03) and OS at 24 months (2.78, 1.12–8.29). Aneuploidy Score was higher in non-responders under both OS thresholds (r = 0.20 and 0.25) and the re-derived interval (r = 0.21), and weakly under platinum status (r = 0.13, q = 0.26). In the same patients, no contrast between platinum status and the OS thresholds in these characteristics could be distinguished from zero, and again only large differences were detectable (S5 Table, with raw P values). Older age was associated with both OS thresholds (r = 0.26 and 0.28) but not with platinum status (r = 0.07), as predicted in advance for the 24-month threshold in patients with RNA-seq data (S1 Text). First-course response showed no clear association with aneuploidy (r = −0.03). These markers therefore do not resolve the comparison between labels.

Exploratory: associations with the platinum-free interval and with survival after progression

When overall survival was divided into the platinum-free interval (408 patients at risk at the end of platinum) and survival after progression (S10 Table), *BRCA1/2* carriers had a longer platinum-free interval (hazard ratio 0.55, 95% CI 0.38–0.79), and *CCNE1* amplification (1.58, 1.19–2.09) and higher Aneuploidy Score (1.23 per SD, 1.10–1.38) a shorter one, with weaker or no associations with survival after progression (Fig 4). Age and residual disease > 1 cm were associated with shorter survival after progression (1.26 per 10 years, 1.12–1.42; and 1.41, 1.03–1.93) but not clearly with the platinum-free interval (1.09, 0.98–1.20; and 1.08, 0.82–1.41). Setting the excluded intervals to 1 day changed no hazard ratio by more than about 0.03 and no conclusion. Any residual disease was associated with both. A clearer association in one model

than in the other is not evidence that the associations differ. Separately, a logistic regression on age, stage and residual disease predicted death before 18 and 24 months (cross-validated AUC 0.64 and 0.62) whereas discrimination was near chance for platinum status (0.48), the re-derived interval (0.52) or the broad combined label (0.49).

Classifiers trained on each label predict modestly

Within TCGA, expression classifiers predicted each label modestly (Fig 3A): mean cross-validated AUC was 0.63 for platinum status (range over models and repeats 0.56–0.69; S6 Table), 0.61 for the re-derived interval, 0.63 for OS at 18 months and 0.68 for OS at 24 months. Models trained on the OS thresholds predicted platinum status with AUCs of 0.55 and 0.60, and models trained on platinum status predicted OS at 24 months with an AUC of 0.63. These all-patients comparisons use different patients and class balances for each label and are descriptive.

In the controlled comparison (134 patients, 39 platinum-resistant, the same folds for every label; Fig 3B), platinum-trained and 24-month-OS-trained models did not differ detectably in predicting platinum status (difference in AUC −0.005; conditional 95% interval −0.105 to +0.087). The corresponding difference for 18-month-OS-trained models was +0.080 (−0.045 to +0.201), but with only 7 positives this comparison is particularly unstable (S6 Table); on the 24-month OS label, no improvement of 24-month-OS-trained over platinum-trained models was demonstrated (difference −0.069, −0.236 to +0.081). The conditional interval reflects the sampling of patients, not the refitting of models. In the separate out-of-bag sensitivity analysis, the median differences were +0.01 (2.5th–97.5th percentiles −0.18 to +0.16), +0.06 (−0.15 to +0.24) and −0.004 (−0.23 to +0.22) for the three contrasts (Fig 3B); each replicate is a single fit evaluated on 49 patients on average (with as few as one positive for the 24-month OS label), so

this range describes the variability of single refits, not a confidence interval for the estimates above. No clear difference between training labels was demonstrated (S1 Text).

On GSE63885, models predicted its platinum-resistance label weakly whichever label they were trained on (mean AUC 0.56–0.63); on its clinical-response label, which has 10 non-responders, models trained on the platinum labels had AUCs of 0.67 and 0.71 and OS-trained models 0.49 and 0.52, a comparison that was not pre-specified, has no interval and is descriptive only (S1 Fig, S7 Table). In the gene-set transfer experiment, the 10 genes of Chen et al. separated platinum-resistant from sensitive TCGA patients better than random gene sets (AUC 0.735) but not GSE63885 patients (0.499), and the 5 genes of Gao et al. did not differ from random sets (S8 Table). Selecting genes on the full dataset before splitting raised mean AUC at the 18- and 24-month thresholds by +0.21 with supervised selection and by +0.006 with unsupervised selection (S2 Fig, S9 Table).

Discussion

Candidate platinum-response labels in TCGA-OV differ in coverage, in which patients they call resistant, and in what they measure. The two platinum labels agree almost perfectly. Platinum status covers just over half of treated patients (277), the re-derived interval 389, and the combined labels about four-fifths (broad) or nearly three-quarters (provenance-filtered). The OS thresholds cover more patients, but at no examined cut-off between 6 and 60 months do they agree better than moderately with platinum status. At 18 and 24 months they flag an early-failure group: most flagged patients with a known status are platinum-resistant, but most resistant patients are not flagged, because platinum-resistant disease is compatible with survival beyond

two years, and half of the flagged patients have no platinum status to check against. At longer thresholds they catch more resistant patients at the cost of flagging sensitive ones.

The exploratory decomposition is consistent with this picture but does not establish it. *BRCA1/2*, *CCNE1* and aneuploidy were associated more clearly with the platinum-free interval, and age and bulky residual disease with survival after progression, but the two models were not compared formally, and clinical characteristics alone predicted the OS labels but not platinum status. An OS-threshold label may therefore mix early platinum failure with other prognostic information; the decomposition raises this possibility but does not test it, and it would need confirmation in another cohort. These marker analyses did not distinguish the labels in this cohort: *BRCA1/2* and *CCNE1* are associated with survival as well as with platinum response [43–45], so they point the same way under every label, and our same-patient comparisons could detect only large differences. First-course response agreed weakly with every other label and showed no clear *BRCA1/2* association, consistent with a measure of response to surgery and chemotherapy together.

Within TCGA, expression predicted each training label with cross-validated AUCs near 0.6–0.7. In the same patients with the same folds, no difference was detected between platinum-trained and 24-month-OS-trained models, so this study does not show that the choice of training label changes classifier performance; given the conditional interval (−0.105 to +0.087) and the wider variation of single refits of the whole pipeline, moderate differences in either direction cannot be excluded. With a standard, untuned pipeline and 41 resistant patients in the all-patients analysis, these estimates are imprecise and describe what this pipeline achieves, not an upper bound for the data. They are nonetheless well below figures reported in several studies [2,25,26,37], which used other cohorts, data types and designs; selecting genes before splitting, one documented

route to higher figures [27–30], inflated AUC by about 0.2 in our pipeline, which shows that it can inflate such figures but not that it explains any particular published result.

Villalobos et al. re-annotated the cohort into platinum-sensitive, -resistant and -refractory groups (212, 91 and 28 patients), with median overall survival of 27.0 months in patients whose treatment failed 1 to 6 months after the last platinum dose and 11.6 months in refractory patients [13]; our resistant category includes both, and the longer median fits our finding that many resistant patients survive beyond the usual OS thresholds. Cocchi et al. released the patient barcodes and platinum status of their three cohorts (230, 201 and 142 patients) with their workflows [14]. Our label table differs from both in its sources, in covering all 489 platinum-treated patients, in recording the provenance and flags of each re-derived call, and in placing platinum, survival and response labels side by side; it does not supersede their annotations. Patients without a platinum status are a mixed group — some died early or never responded, others were followed too briefly — so the missing values are unlikely to be random. Re-deriving the interval recovers about half of them (117 of 212), and where both labels exist they agree closely, which shows consistency with the published reference rather than accuracy in the patients without it. Two-thirds of the added calls (78) rest on unflagged records, and the provenance-filtered label keeps only those.

We recommend the following for TCGA-OV studies of platinum response. State the label, its definition, its source field and the number of patients in each class. Prefer a platinum-specific label. If platinum status is extended with the interval re-derived from TCGA-CDR, document the provenance of each added call and report results with and without the added patients; the provenance-filtered candidate extension we release (S11 Table) is one such option, not a label shown to be more accurate, and its filter also removes borderline patients whose records may be

correct. Report any OS threshold as a survival end-point, not a platinum-response label, and expect low sensitivity for platinum resistance at short thresholds and poor specificity at long ones. Do not relabel first-course response as platinum resistance. Within cross-validation, estimate feature selection and preprocessing from training data only, and describe separately any normalisation that uses an external cohort's predictor distribution.

Limitations

The label analyses use one resource (TCGA-OV), with GSE63885 used only for external transfer, and platinum status is available for only 145 patients with RNA-seq data (41 resistant), which limits the precision of every classifier result. The same-patient label comparisons were underpowered and do not establish equivalence, and the controlled classifier comparison, whose interval is conditional on the fitted models, cannot exclude a moderate difference in either direction. Platinum status depends on the completeness of treatment and follow-up records, the re-derived interval shares TCGA's progression dates and counts death with tumour as progression, and their close agreement shows consistency rather than independent truth. The provenance audit checks the records behind each call, not the clinical events, and at least two disagreements with unflagged records remain unexplained; the 150–215-day flag may make the filtered population less representative. Sensitivity and PPV of the OS thresholds are conditional on patients having a platinum status, and the bounds we report are wide. The cohort excludes 89 patients without a treatment record, whose survival was poorer (median 27.6 months, against 47.7 in the cohort) and only five of whom had a platinum status; the finding that patients without a status did not have shorter survival holds only within the defined cohort. Both platinum labels use the six-month dichotomy that consensus has replaced for trial design [5], and neither separates platinum-refractory from platinum-resistant disease. Excluding patients censored

before an OS threshold can bias threshold labels [21]; weighting changed the estimates little but rests on uncheckable assumptions about censoring. *BRCA1/2* carriers were limited to truncating and splice-site variants called in the 2011 release, whose sequenced patients are a subset of the cohort, not a random sample of it; missense variants of uncertain significance and homologous-recombination-deficiency scores were not assessed. The decomposition, clinical-only models, threshold sweep and reference bounds were exploratory and added after the main results were known, and the decomposition describes patients event-free and under observation at the end of first-line platinum, not the whole cohort. Several pre-specified predictions failed, including that platinum-trained classifiers would predict platinum status better than OS-trained ones (S1 Text). TCGA expression data were processed centrally before any split we made, so upstream batch effects cannot be controlled by our pipeline.

Conclusion

In TCGA-OV, platinum-specific labels are consistent but cover only part of the cohort, and the re-derived interval extends them with records that can be checked patient by patient. Overall-survival thresholds, evaluated here as candidate surrogates, cover more of the cohort but agree at best moderately with platinum status. Expression predicts each of the four training labels modestly, and in the same patients the training label made no detectable difference to classifier performance. Which label a classifier is trained on determines how many patients it learns from and what it is asked to find; stating that choice, with class sizes, is a precondition for comparing models of platinum response built on these data. The versioned patient-level label table released with this paper (S11 Table; Zenodo, https://doi.org/10.5281/zenodo.20802753) gives each patient’s labels and the provenance of each re-derived call.

Data and code availability

TCGA data were obtained from cBioPortal (ov_tcga_pan_can_atlas_2018, ov_tcga_pub) and the TCGA-CDR supplementary table [6] (Genomic Data Commons file 1b5f413e-a8d1-4d10-92eb-7c4ae739ed81). GSE63885 is available from the Gene Expression Omnibus. All code, analysis outputs, pre-specified predictions and figures are available at https://github.com/AaronC-bioinfo/ovarian-cancer-platinum-response and are archived at Zenodo (https://doi.org/10.5281/zenodo.20802753). The patient-level label table (version 1.0, with a data dictionary) is provided as S11 Table and in the Zenodo record.

Author contributions

AMAC: conceptualization, data curation, formal analysis, investigation, methodology, project administration, software, validation, visualization, writing – original draft, writing – review & editing.

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Competing interests

The author has declared that no competing interests exist.

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Supporting information

S1 Fig. Transfer of TCGA-trained classifiers to GSE63885, by training label.

S2 Fig. Feature-selection leakage: AUC inflation from supervised and unsupervised gene selection before data splitting.

S1 Table. Full literature extraction for Table 1, with verbatim quotations and their page locations.

S2 Table. Pairwise agreement and OS-threshold test characteristics: full cohort, patients with RNA-seq data, against the combined label, at every threshold from 6 to 60 months, and with inverse-probability-of-censoring weights.

S3 Table. *BRCA1/2* and *CCNE1* associations for every carrier definition (primary, any non-silent variant, germline only, somatic only, *BRCA1* and *BRCA2* separately), with same-patient differences between labels and minimum detectable differences.

S4 Table. Cohort characteristics, cohort flow, and patients with and without a determinate platinum status, with bounds under the missing reference standard.

S5 Table. Associations with clinical and tumour characteristics, with raw P values and q values, in all determinate patients and in the same patients.

S6 Table. Within-TCGA cross-label AUC for each model and repeat.

S7 Table. Transfer to GSE63885: AUC with 95% bootstrap intervals for every training label, model and GEO label.

S8 Table. Gene-set transfer experiment: published gene sets as fixed inputs, with the random-gene-set null distribution.

S9 Table. Feature-selection leakage at the 12-, 18- and 24-month thresholds.

S10 Table. Exploratory analyses: decomposition of overall survival (platinum-free interval in patients at risk at the end of platinum, with the earlier model that set short intervals to 1 day as a sensitivity analysis), clinical-only models and the redaction sensitivity analysis; Cox models of the progression-free interval.

S11 Table. Patient-level label table (version 1.0): every label, the provenance-filtered and broad combined labels, provenance class, event source and flags for the 489 platinum-treated patients, with a data dictionary, class counts, the four discordant patients and the treatment-anchor sensitivity analysis.

S1 Text. Pre-specified predictions and decision rules for each analysis, with observed outcomes, and the analysis history and deviations.

S2 Text. Reporting checklists (STROBE; STARD 2015 for the OS-threshold analysis).

Figure legends

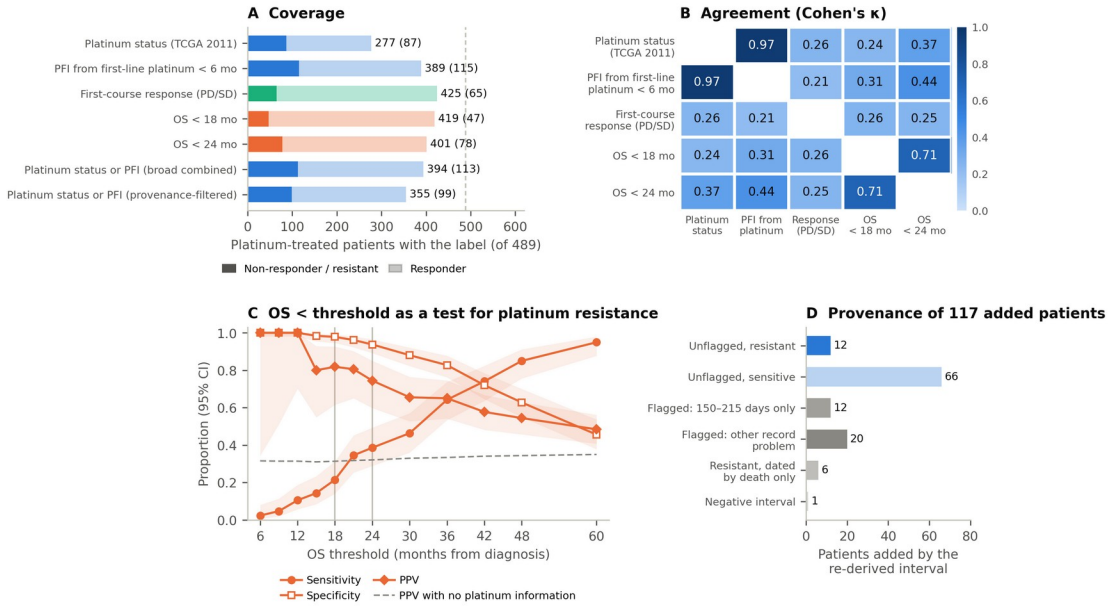


Fig 1. Coverage and agreement of candidate labels in 489 platinum-treated TCGA-OV patients. (A) Number of patients in whom each label is determinate; dark segment, non-responders or resistant; light segment, responders; dashed line, all 489 patients; colour denotes label family. Numbers give patients labelled (non-responders or resistant). The broad combined label uses platinum status where available and the re-derived platinum-free interval otherwise; the provenance-filtered combined label (preferred under the pre-specified rule) adds re-derived calls only for unflagged patients. (B) Cohen's κ for each pair of labels, in patients with both labels determinate. (C) Sensitivity, specificity and positive predictive value (PPV) of death before each threshold as a test for platinum resistance, with TCGA platinum status as the reference; shaded bands, Wilson 95% confidence intervals; dashed line, PPV expected if the OS label carried no information about platinum status; vertical lines, 18 and 24 months. (D) Provenance of the 117 patients added by the re-derived interval, each counted once (in the order negative interval, resistant call dated only by a death, other record problem, interval within 150–215 days only): unflagged resistant or sensitive; flagged only for an interval within 150–215 days; flagged for another record problem (redacted TCGA-CDR record, adjuvant platinum without a stop date, or a later-line regimen before the event, with or without a near-threshold interval); resistant call dated only by a death counted as progression; or negative interval.

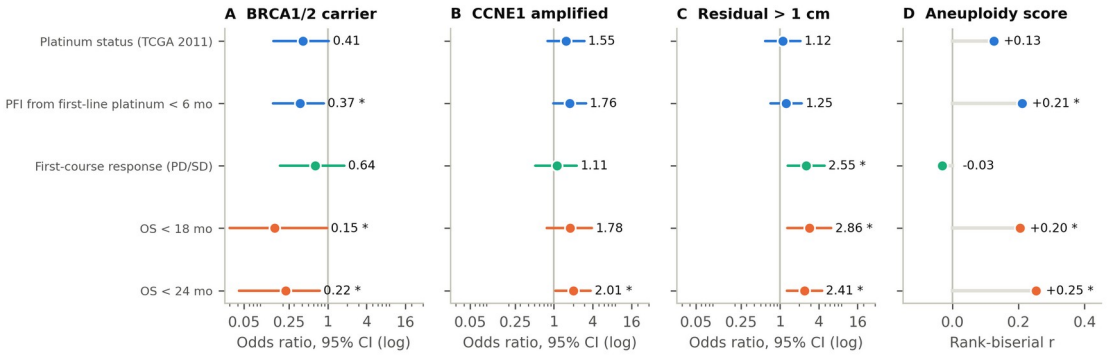


Fig 2. Characteristics associated with platinum response and prognosis under each label. Odds ratio (95% CI, log scale) for being labelled a non-responder in (A) carriers of a germline or somatic truncating or splice-site *BRCA1/2* variant (272 sequenced patients), (B) tumours with *CCNE1* amplification and (C) patients with residual disease > 1 cm. (D) Rank-biserial correlation between Aneuploidy Score and non-response (positive, higher in non-responders). Each label uses all platinum-treated patients in whom it is determinate. A–C: * 95% CI excludes 1; lower bounds below 0.03 are drawn at 0.03. D: * Benjamini–Hochberg q < 0.05 within label (five characteristics). Colour denotes label family: platinum labels (blue), first-course response (teal), OS thresholds (orange).

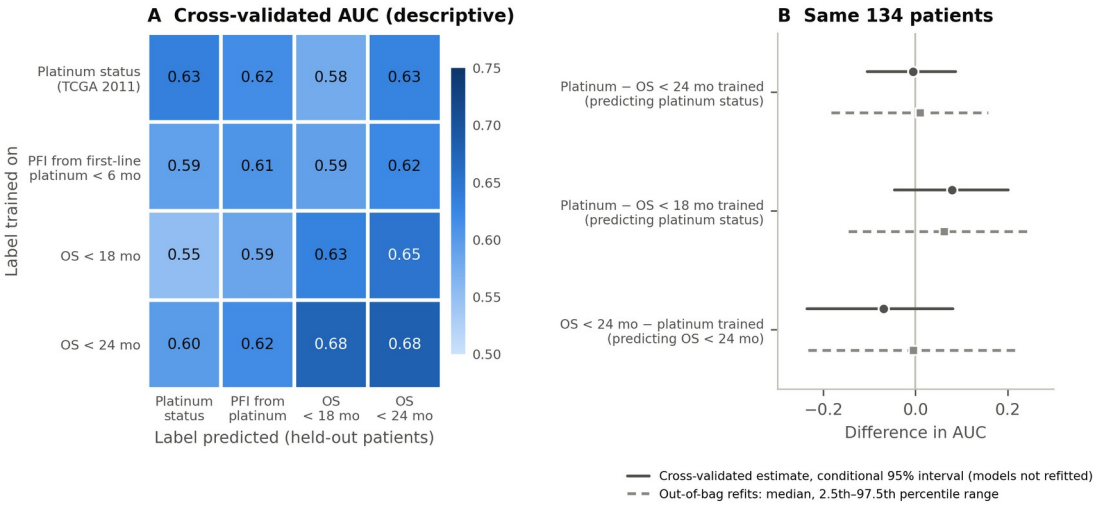


Fig 3. What classifiers trained on each label predict within TCGA. (A) Mean cross-validated AUC (three models, five repeats of five-fold cross-validation, 251 patients with RNA-seq data) for each training label (rows) against each label in held-out patients (columns); descriptive, because each label has its own patients and class balance. (B) Controlled comparison in the same 134 patients: difference in AUC between training labels (mean of three models). For each contrast, the upper row (circles, solid lines) gives the cross-validated estimate (same folds for

every label) with its conditional 95% interval from resampling patients without refitting the models; the lower row (squares, dashed lines) gives the median and 2.5th–97.5th percentile range of out-of-bag differences from 500 bootstrap refits of the whole pipeline, a sensitivity analysis of a different quantity.

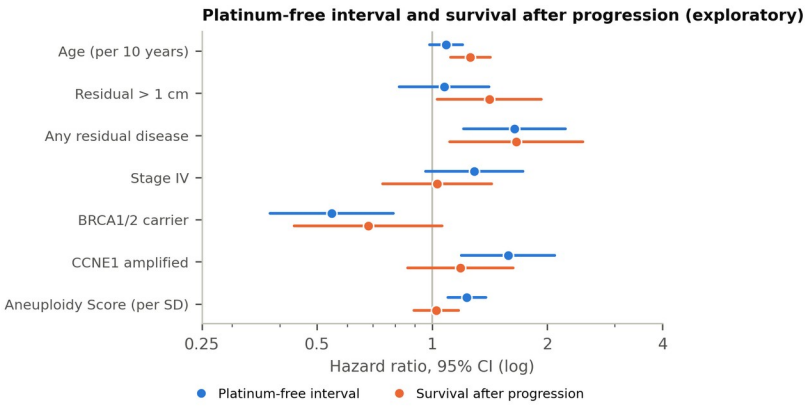


Fig 4. Associations with the platinum-free interval and with survival after progression (exploratory). Hazard ratios (95% CI, log scale) from univariable Cox models of the platinum-free interval from the end of first-line platinum (blue; 408 patients at risk at the end of platinum, with an interval above 0 days) and of survival after progression among patients who progressed (orange; patients whose progression event was their death excluded). Times from TCGA-CDR. The two models were not compared formally.

Tables

751 **Table 1. Labels used by studies of expression and platinum response in TCGA-OV and related cohorts.**

Study	TCGA-OV label and stated definition	TCGA n (resistant / sensitive)	External or other cohorts	Design	Reported discrimination
Gonzalez Bosquet 2016 [31]	Complete response, progression-free 6 months after first platinum-based treatment, vs refractory or recurrence within 6 months of completion	158 / 292	Six GEO/EBI cohorts recoded to the same classes	34 genes selected on the full TCGA set; internal cross-validation type not stated	AUC 0.64–0.79 (TCGA), 0.45–0.73 (external)
Sun J 2019 [32]	“Chemo-response information”; definition not stated	Not stated (TCGA n = 557, microarray)	Several GEO cohorts	Rank-based score, cut-off 0	No AUC
Choi 2020 [33]	Recurrence < 6 months vs > 12 months after last	110 / 160	GSE9891 (survival only)	No classifier	Not applicable

	primary chemotherapy; 6–12 months excluded				
Gonzalez Bosquet 2021 [34]	Progression-free ≥ 6 months after first platinum-based treatment; ≥ 6 months of follow-up required	149 / 189 (validation set)	Trained on 88 patients from one institution	Clinical and multi-omic lasso models; TCGA used for validation	TCGA AUCs not reported numerically in the main text or Fig 4
Sun X 2021 [35]	No response label (progression-free survival used as surrogate)	—	GSE131978, 7 / 4; GSE51373, 10 / 13	No classifier	Not applicable
Guo & Li 2021 [36]	Relapse within 6 months after completing chemotherapy	49 / 205	—	Single-gene ROC analysis	Per-gene AUC 0.58–0.60
Chen 2022 [25]	Platinum-free interval < 6 months from end of first chemotherapy course;	69 / 161	GSE63885, 34 / 41	Genes selected on the full labelled set; no held-out TCGA split; SVM with	AUC 0.971 (TCGA), 0.926 (GSE63885)

	TCGA field not stated			100-fold CV	
Li 2022 [37]	“Known platinum sensitivity” from cBioPortal; definition not stated	90 / 197 (supplement; equal to TCGA 2011 platinum status)	GSE15622, 6 / 9	Held-out split and CV not stated	AUC 0.841 (TCGA), 0.796 (GSE15622)
Wang C 2024 [38]	TCGA label definition not stated (counts equal to TCGA 2011 platinum status); relapse within six months stated for the external ROC-plotter analysis	90 / 197	Public cohorts via ROC-plotter	Single-gene ROC analysis	Per-gene AUC > 0.6
Gao 2025 [26]	Source of bulk TCGA labels not stated (interval < 6 months stated for single-	287 TCGA in a pooled 404 from four	GSE63885, n = 75	Random 8:2 split of TCGA, GSE15622 and GSE51373 (GSE63885	AUC 0.993 (test), 0.989 (GSE63885)

	cell cohort)	datasets, including the GSE63885 validation set (pooled counts as printed: 71 sensitive / 333 resistant)		held out); five-fold CV for hyperparameters	
Kilim 2025 [39]	Refractory (progression or stable disease within six cycles) vs no progression within 2 years; proteomics and histology, not mRNA	127 patients; classes not stated	PTRC-HGSOC	Train on one cohort, test on the other; five-fold CV	AUC 0.752 (PTRC → TCGA), 0.835 (TCGA → PTRC), 0.613 (TCGA five-fold CV); multimodal, primary tumours
Wang C 2025 [40]	No TCGA response-classifier cohort; GEO	—	Merged GSE114206,	Selection on the merged	Per-gene AUC 0.66–0.72

labels, definition not stated	GSE51373, GSE63885; GSE30161 (test)	set; no held-out split	(merged), 0.55–0.75 (GSE30161)

Table 2. Candidate labels in platinum-treated TCGA-OV patients.

Label	Definition (1 = non-responder)	Source	Platinum cohort, n (1 / 0)	With RNA-seq, n (1 / 0)
Platinum status	Platinum-free interval < 6 months from last primary platinum dose, with progression	TCGA 2011 [1]	277 (87 / 190)	145 (41 / 104)
Platinum-free interval	Progression < 183 days after end of first-line platinum	TCGA-CDR + treatment timeline	389 (115 / 274)	188 (53 / 135)
Provenance-filtered combined label (candidate extension)	Platinum status where determinate, otherwise platinum-free interval in patients with an unflagged record	As above, with provenance audit	355 (99 / 256)	167 (43 / 124)
Broad combined label (sensitivity analysis)	Platinum status where determinate, otherwise platinum-free interval	As above	394 (113 / 281)	189 (51 / 138)
First-course	Stable or progressive vs	TCGA-CDR,	425 (65 /	219 (35 /

response	complete or partial response	then TCGA 2011	360)	184)
OS < 18 months	Death before 18 months vs survival beyond	PanCancer Atlas clinical	419 (47 / 372)	206 (22 / 184)
OS < 24 months	Death before 24 months vs survival beyond	PanCancer Atlas clinical	401 (78 / 323)	199 (36 / 163)

753 **Table 3. Association of each label with characteristics linked to platinum response and**
754 **prognosis.**

Characteristic	Platinum status	Platinum-free interval	First-course response	OS < 18 months	OS < 24 months
BRCA1/2 carrier, OR	0.41 (0.14–1.04)	0.37 (0.14–0.87)†	0.64 (0.18–1.80)	0.15 (0.004–0.97)†	0.22 (0.04–0.75)†
CCNE1 amplified, OR	1.55 (0.80–2.97)	1.76 (0.98–3.15)	1.11 (0.52–2.26)	1.78 (0.78–3.89)	2.01 (1.06–3.75)†
Residual > 1 cm, OR	1.12 (0.59–2.09)	1.25 (0.71–2.18)	2.55 (1.31–4.91)*	2.86 (1.30–6.29)*	2.41 (1.28–4.50)*
Any residual, OR	3.22 (1.40–8.35)†	3.04 (1.44–7.03)†	1.54 (0.67–3.99)	2.93 (0.87–15.43)	2.78 (1.12–8.29)†
Stage IV, OR	1.02 (0.46–2.17)	1.09 (0.57–2.03)	1.25 (0.57–2.57)	1.75 (0.76–3.79)	1.55 (0.78–2.97)

Aneuploidy Score, r	0.13	0.21*	−0.03	0.20*	0.25*
TMB, r	−0.16	−0.06	−0.03	−0.08	−0.08
Age, r	0.07	0.15	0.06	0.26*	0.28*

OR, conditional odds ratio for non-response (95% CI); r, rank-biserial correlation (positive = higher in non-responders). Each label uses all platinum-treated patients in whom it is determinate; *BRCA1/2* and *CCNE1* in sequenced and copy-number-profiled patients. Any residual versus none was an alternative coding added after the main results were known. † 95% CI excludes 1. * Benjamini–Hochberg $q < 0.05$ within label across the five pre-specified clinical and tumour characteristics.